Quiz — 1.2.4 Types of Programming Language

Name: ____ Date: _ ***Mark:*** ___ / 25

OCR H446 · Component 01 · 1.2.4

Section A — Multiple choice (8 marks)

1. A *programming paradigm* is best described as: A. A specific programming language B. A style or approach to writing programs C. A type of CPU D. A piece of utility software Your answer: ☐

2. The **procedural** paradigm organises a program using: A. Classes and objects only B. Sequence, selection and iteration in procedures/subroutines C. Logic facts and rules D. Spreadsheets and macros Your answer: ☐

3. Assembly language uses **mnemonics** such as `ADD` and `STA` and is: A. A high-level, machine-independent language B. Processor-specific, with ≈ one mnemonic per machine instruction C. Only usable inside a web browser D. Identical on every processor Your answer: ☐

4. In the LMC, the instruction `BRZ xx` branches to address xx only if the Accumulator is: A. Greater than zero B. Equal to zero C. Less than zero D. Negative Your answer: ☐

5. In **immediate** addressing, the operand holds: A. The actual value to be used B. The address of the value C. The address of a location holding the address of the value D. A base address plus an index Your answer: ☐

6. **Indirect** addressing means the operand is: A. The actual value B. The address holding the value C. The address of a location that holds the address of the value D. A value added to the index register Your answer: ☐

7. Which OOP concept means a subclass acquires the attributes and methods of a superclass? A. Encapsulation B. Inheritance C. Polymorphism D. Instantiation Your answer: ☐

8. An **object** is best defined as: A. A template defining attributes and methods B. A specific instance of a class C. A subroutine belonging to a class D. A private variable of a class Your answer: ☐

Section B — Short answer (12 marks)

9. Define the OOP terms **class**, **attribute** and **method**. **[3]**

10. Explain **encapsulation** and state one benefit it provides. **[2]**

11. Addressing-mode calculation. A processor executes `LDA 6`. The relevant memory contents are:

> Address 6 holds the value **9** · Address 9 holds the value **4** · The index register holds **2**.

State the value loaded into the Accumulator for each mode. **[3]**

(a) Immediate: __ **(b) Direct: _ *(c) Indirect:* ___**

(d) Indexed (value taken from the resulting address): _______ *(state which address is used)*

12. Explain **two** reasons a programmer might choose to write part of a program in assembly language rather than a high-level language. **[2]**

13. Explain what **polymorphism** is and give a short example of how it could be used. **[2]**

Section C — Exam-style question (5 marks)

14. A team is writing a program that models different types of vehicle (Car, Bus, Motorbike), all of which share common data such as `speed` and behaviour such as `move()`, but each displays itself differently.

Explain how **inheritance** and **polymorphism** could be used in this object-oriented program, and give **one** benefit of designing it this way. **[5]**